

Is this project *real*, bankable, and worth building?

Waste-to-Energy plant — illustrative sample report

FEASIBILITY VERDICT



Conditionally feasible — proceed with defined conditions

The plant is physically sound and the energy balance closes. Bankability is gated by feedstock-contract security and a tightened gate-fee assumption. Two conditions must clear before capital commitment.

Prepared for	[Client] — illustrative
Project	12 MW _{th} RDF waste-to-energy facility
Location	Coastal municipal site, Northern Europe
Study class	AACE Class 4 (feasibility / screening)
Turnaround	10 business days
Prepared by	Aidevo AS, Oslo — physics-native feasibility
Engine	W2E v-sample · validated by EIGEN v2.1

Advisory only — not for final investment decision. AACE Class 4 accuracy (-30% / +50%). Figures illustrative for format demonstration.

One decision, on one page

This study evaluates a 12 MW_{th} refuse-derived-fuel (RDF) waste-to-energy facility against four reality tests: is it physically possible, is it engineerable, is it economic, and is it bankable. The verdict is **Class C — conditionally feasible**. The thermodynamics close and the technology is mature; the project's viability turns on two commercial conditions, not on physics.

C DECISION CLASS	€186/t BREAK-EVEN GATE FEE	11.4% PROJECT IRR (BASE)
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The one decision: Do not commit capital until (1) a binding 10-year feedstock supply contract is secured at ≤€155/t variability, and (2) the gate-fee assumption is re-based on the signed offtake rather than the regional average. With both conditions met, the project moves to Class B and is bankable.

GREEN — Physics & energy balance. Mass and energy balance close within 2.1%. Net electrical output 9.8 MW _e , parasitic load 18%. No thermodynamic showstopper.
YELLOW — Gate-fee assumption. Base case uses €170/t regional average; signed contracts in this market cluster at €140–155/t. The €15–30/t gap moves IRR by ±2.6 points.
RED — Feedstock security. No binding long-term supply contract. Merchant feedstock exposure makes the financing case fragile; lenders will require contracted tonnage before debt.

Where the project is strong, and where it bends

Each axis is scored 0–1 by the Noema epistemic engine. An axis below 0.60 is a binding condition, not a soft warning. Five of six axes pass; Economic and Market sit at the threshold.



Reading the vector: Physical (0.91) and Engineering (0.84) confirm the plant works as designed. Market (0.58) falls below threshold — the feedstock-supply market is the binding constraint, consistent with the RED flag. Economic (0.62) is marginal and sensitive to the gate-fee re-basing. Regulatory and Temporal are comfortable.

CONTRADICTION GRAPH — FLAGGED ASSUMPTIONS

ASSUMPTION	CONFLICTS WITH	SEVERITY
Gate fee €170/t (base)	Signed-contract cluster €140–155/t	Medium
Merchant feedstock supply	Lender debt-sizing requirement	High
92% availability yr 1	Commissioning ramp for RDF grate	Low

Mass & energy balance, physics-native

The facility processes RDF on a moving-grate combustor with steam-Rankine power recovery. All figures are produced by the W2E engine and cross-checked against EIGEN's thermodynamic constraint pack (steam tables, combustion stoichiometry, Carnot ceiling).

PARAMETER	VALUE	BASIS
RDF throughput	14.2 t/h	Design feed rate
RDF lower heating value	14.8 MJ/kg	EN 15359, sampled
Thermal input	58.3 MW _{th}	Throughput × LHV
Boiler efficiency	84%	Grate + economizer
Gross electrical output	12.0 MW _e	Steam-Rankine, 540°C/45 bar
Parasitic load	2.2 MW _e	Fans, pumps, APC
Net electrical output	9.8 MW _e	Gross – parasitic
Balance closure	±2.1%	EIGEN cross-check

Constraint check: the proposed 32% net electrical efficiency sits below the Carnot ceiling for the cycle (no veto). Flue-gas treatment train (SNCR + bag filter + activated carbon) is sized for the EU Industrial Emissions Directive limits. No hard physical veto raised.

ECONOMICS (AACE CLASS 4)

LINE	VALUE	NOTE
Total installed CAPEX	€68.4M	-30% / +50%
Annual OPEX	€7.1M/yr	incl. APC consumables
Gate fee revenue (base)	€19.3M/yr	@ €170/t, 113 kt/yr
Power revenue	€6.2M/yr	@ €72/MWh
Project IRR (base)	11.4%	20-yr, 70% debt
IRR (gate fee €150/t)	8.8%	re-based case

What to do next — in order

This study is a decision instrument, not a 200-page binder. The path from Class C to bankable Class B is two conditions and three actions.

- **Secure binding feedstock supply.** Negotiate a 10-year contracted-tonnage agreement at $\leq \text{€}155/\text{t}$. This clears the RED flag and the Market axis. Without it, no lender will size debt.
- **Re-base the gate-fee assumption.** Replace the $\text{€}170/\text{t}$ regional average with the signed-offtake number. Re-run the economic case; confirm IRR holds above the hurdle.
- **Commission a Pre-FEED study** (Aidevo Pre-FEED package) once the two conditions clear — to take CAPEX from Class 4 to Class 3 and produce the investor-ready package.

If both conditions clear: project moves to Class B (bankable). Estimated re-base IRR with contracted feedstock at $\text{€}150/\text{t}$ and gate fee re-based: 10.9% — above a 9% hurdle.

PROVENANCE & VALIDATION

Every quantitative claim in this report was generated by the Aidevo W2E engine and validated against EIGEN's physics constraint packs. Claims that could not be falsified are admissible; claims violating a hard physical constraint are vetoed and never appear. This report is cryptographically signed; the signature below binds this exact content.

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EIGEN-admissibility: PASS (0 hard vetoes, 3 flagged assumptions)
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Aidevo AS · Oslo, Norway · Physics-native industrial feasibility. This is an illustrative sample demonstrating report format and depth. All figures are synthetic. Advisory only; not for final investment decision. AACE Class 4 accuracy.